

1/10/26

Ex: 12 End-End Communication at Transport layer.

a) Implement echo client-server using TCP/UDP sockets.

TCPimport socket
import threading

... SERVER ...

def start_server():

server = socket.socket(socket.AF_INET, socket.

SOCK_STREAM)
server.bind(('127.0.0.1', 5000))

server.listen(1)

print(f"[SERVER] Listening on 127.0.0.1:5000")

while True:

data = conn.recv(1024).decode()

if not data or data.lower() == 'quit':

print(f"[SERVER] Connection closed by client.")

break

print(f"[SERVER] Received: {data}")

conn.send(f"Server received: {data}".encode())

conn.close()

server.close()

... client ...

def start_client():

client = socket.socket(socket.AF_INET, socket.

SOCK_STREAM)
client.connect(('127.0.0.1', 5000))

print(f"[CLIENT] Connected to Server")

while True:

msg = input("Enter message (or 'quit' to exit): ")

client.send(msg.encode())

if msg.lower() == 'quit':

break

```
response = client.recv(1024).decode()
print(f"[SERVER RESPONSE] {response}")
client.close()
```

```
if __name__ == "__main__":
```

```
    server_thread = threading.Thread(target=start_server,
```

```
                                     daemon=True)
    server_thread.start()
```

```
    start_client()
```

O/p:

[SERVER] Received : hello

[SERVER RESPONSE] Server Received : hello

Enter message (or 'quit' to end): ~~Quit~~

UDP:

```
import socket
```

```
import threading
```

```
def udp-server():
```

```
    s = socket.socket(socket.AF_INET, socket.
```

```
                        SOCK_DGRAM)
```

```
    s = bind(('127.0.0.1', 12000))
```

```
    print("[SERVER] Running on 127.0.0.1:12000")
```

```
    while True:
```

```
        data, addr = s.recvfrom(1024)
```

```
        msg = data.decode()
```

```
        if msg.lower() == 'quit':
```

```
            print(f"[SERVER] client {addr} ended the connection.")
```

```
            break
```

```
            print(f"[SERVER] Received from {addr}: {msg}")
```

```
            s.sendto(data, addr)
```

```
    s.close()
```

```
    print("[SERVER] stopped.")
```

```
# --- client ---
```

```
def udp-client():
```

```
    c = socket.socket(socket.AF_INET, socket.
```

```
                        SOCK_DGRAM)
```

```
    server = ('127.0.0.1', 12000)
```

```
    while True:
```

```
        msg = input("Enter message (or 'quit' to stop): ")
```

```
        c.sendto(msg.encode(), server)
```

```
        if msg.lower() == 'quit':
```

```
            print("[CLIENT] closing connection.")
```

```
            break
```

```

data, _ = c.recvfrom(1024)
print("Echo from Server:", data.decode())

c.close()
print("[CLIENT] Stopped.")

# --- MAIN ---
if __name__ == "__main__":
    threading.Thread(target=udp_server,
                      daemon=True).start()
    udp_client()

```

Output

```

Enter message (or 'quit' to stop): Hi
[SERVER] Received from ('127.0.0.1', 62859): Hi
Echo from Server: Hi
Enter message (or 'quit' to stop): quit.

```

It

Result:

Thus, the client and server side program for TCP and UDP was written and executed successfully.